

Project ARIA: A Cognitive Multi-Agent Vision-Language-Action Architecture for Autonomous Manipulation on Low-Cost 5-DoF Robotic Arms

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Abstract—Autonomous robotic manipulation has experienced transformative advances driven by Vision-Language-Action (VLA) foundation models. However, contemporary robotic foundation policies are monolithic black boxes demanding datacenter-scale compute (> 24 GB VRAM), incurring prohibitive inference latencies (> 100 ms), and failing catastrophically on low-cost underactuated manipulators. In this paper, we present the research formulation, architectural implementation, and empirical validation of Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent VLA architecture specifically engineered to deliver high-precision physical autonomy on budget 5-DoF manipulators within an 8.0 GB VRAM budget. ARIA systematically resolves core robotic bottlenecks through: (1) a Zero-Cost-Depth perception pipeline combining YOLOv8m, Segment Anything 2 (SAM2), and Depth-Anything v2, recovering sub-centimeter metric 3D scene geometry ($RMSE = 8.2$ mm) from dual commodity 2D RGB cameras via kinematic claw grounding; (2) a closed-form analytical Inverse Kinematics (IK) solver resolving the 5-DoF anthropomorphic chain in 0.08 ms ($231\times$ faster than numerical solvers) with 100% reachability determinism; (3) a fifteen-agent ROS 2 lifecycle-managed coordination fabric with human-in-the-loop (HITL) safety and emoji-annotated Chain-of-Thought (CoT) telemetry; and (4) real-time feedforward velocity matching ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt} = 0.05$ m/s) for dynamic moving conveyor sorting and underactuated in-hand pivoting. Benchmarked across standardized industrial tasks, ARIA achieves an 89.0% manipulation success rate, outperforming LeRobot ACT (72.0%), OpenVLA (68.0%), and Physical Intelligence π_0 (61.0%) while slashing overall hardware costs by over 95%.

Index Terms—Cognitive robotics, multi-agent systems, vision-language-action (VLA), low-cost manipulators, zero-cost depth estimation, inverse kinematics, conveyor tracking.

I. Introduction

THE emergence of multimodal foundation models and Vision-Language-Action (VLA) architectures has opened unprecedented possibilities for embodied physical artificial intelligence [1], [2]. Contemporary robotic foundation policies—such as RT-2 [3], OpenVLA [2], Octo [4], and Physical Intelligence π_0 [5]—seek to unify

semantic reasoning and low-level continuous motor control within monolithic neural networks.

Despite their impressive zero-shot semantic capabilities, existing foundation policies exhibit severe fundamental limitations when applied to practical, cost-sensitive automation:

- 1) **Prohibitive Computational Footprint:** Monolithic policies range from 3B to 55B parameters, demanding multi-GPU servers (> 24 –80 GB VRAM) and preventing edge deployment.
- 2) **Inference Latency Bottlenecks:** Autoregressive action token generation requires 100–400 ms per step, violating real-time control constraints in dynamic physical settings.
- 3) **Black-Box Inscrutability:** When manipulation fails, monolithic policies provide zero causal visibility into whether failure stemmed from perceptual hallucination, kinematic singularity, or gripper slippage.
- 4) **High Hardware Cost Barriers:** Prior manipulation research relies on high-precision 7-DoF arms (\$25,000–\$60,000) and active structured-light RGB-D cameras (\$500–\$4,000) that exhibit blind zones below 0.2 m and fail on reflective surfaces [6].

Conversely, low-cost educational and light-industrial 5-DoF manipulators ($< \$150$) powered by PWM/serial bus servos are widely available but suffer from physical underactuation, joint elasticity, mechanical backlash, and lack of joint torque sensing. Numerical Inverse Kinematics (IK) solvers (e.g., KDL, TRAC-IK, RTB-LM) frequently diverge on 5-DoF chains, requiring 10–20 ms and failing near kinematic boundaries.

To overcome this divide, we present Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent VLA architecture operating within an 8.0 GB VRAM budget on a consumer laptop.

A. Objectives

The core scientific and engineering goals of Project ARIA are defined across four explicit, measurable objectives:

- 1) **Objective 1: Zero-Cost Metric Depth Perception:** Formulate and validate a markerless, sensor-free

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- 3D spatial perception pipeline using dual commodity 2D RGB streams. By combining overhead YOLOv8m/SAM2 with wrist-mounted Depth-Anything v2 grounded via end-effector claw physical geometry ($Z_{\text{tips}} = 0.065\text{ m}$), achieve sub-centimeter ($RMSE < 10.0\text{ mm}$) metric localization accuracy across the manipulator's workspace without active infrared depth sensors.
- 2) Objective 2: Sub-Millisecond Closed-Form Kinematics: Derive, implement, and benchmark an exact closed-form analytical Inverse Kinematics solver resolving the 2R planar subproblem for 5-DoF articulated arms in $< 0.1\text{ ms}$ ($> 100\times$ faster than numerical optimization routines) with guaranteed 100% reachability determinism and zero singularity divergence.
 - 3) Objective 3: Decentralized Multi-Agent Coordination & Explainable HITL Planning: Architect a fifteen-agent ROS 2 lifecycle-managed coordination system with an asynchronous State Bus, integrating quantized local LLM reasoning (Mistral-7B/Llama-3 via Ollama), real-time emoji-annotated Chain-of-Thought (CoT) telemetry, and human-in-the-loop (HITL) safety confirmation within an 8.0 GB VRAM envelope.
 - 4) Objective 4: Dynamic Moving-Conveyor Tracking & Dexterous Industrial Sorting: Synthesize predictive rendezvous trajectories ($\mathbf{p}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}t$) with feedforward velocity matching ($\mathbf{v}_{\text{ee}} \approx \mathbf{v}_{\text{belt}} = 0.05\text{ m/s}$) to achieve continuous industrial sorting into a 4-pocket palletizing tray ($RMSE < 1.5\text{ mm}$) and defect scrap bin, benchmarked against state-of-the-art foundation VLA models.
- embodiments, yet requiring massive datacenter GPU infrastructure.
- 5) Wu et al. (2024) – GELLO [7]: Demonstrated that low-cost passive teleoperation replica devices can collect high-quality demonstration datasets efficiently for imitation learning.
 - 6) Yang et al. (2024) – Depth-Anything v2 [8]: Established state-of-the-art monocular relative depth estimation using synthetic data mixtures, yet outputting unscaled relative disparity maps lacking metric scale.
 - 7) ClearDepth Team (2024) – ClearDepth [6]: Investigated monocular depth completion for specular and transparent glassware, highlighting fundamental optical limitations of active infrared sensors.
 - 8) Ravi et al. (2024) – Segment Anything Model 2 (SAM2) [9]: Advanced promptable zero-shot video and image segmentation, enabling real-time object tracking across dynamic robotic scenes.
 - 9) Sundaralingam et al. (2024) – CuRobo [10]: Proposed GPU-accelerated numerical trajectory optimization, demonstrating substantial speedups on 7-DoF arms but remaining ill-conditioned on under-actuated 5-DoF kinematics.
 - 10) Wagaa and Kallel (2023) – Analytical IK Formulations [11]: Proved that algebraic closed-form solutions eliminate local minima and boundary singularities inherent in numerical Jacobian inversion.
 - 11) Aradi (2025) – Predictive Industrial Monitoring [12]: Applied streaming machine learning to acoustic and sensor telemetry for automated predictive maintenance on industrial workcells.
 - 12) Pang et al. (2025) – High-Stiffness Manipulation [13]: Investigated trajectory stiffness and contact force control for precision placement in automated manufacturing.

II. Literature Review & Research Gap Identified

A. Literature Review (Recent Research: 2023–2026)

Robotic manipulation research has advanced across several distinct frontiers over the past three years. We synthesize 12 representative recent contributions:

- 1) Kim et al. (2024) – OpenVLA [2]: Developed an open-source 7B parameter VLA foundation model fine-tuned on the Open X-Embodiment dataset. Demonstrates strong generalization across multi-robot setups but requires $> 16\text{ GB}$ VRAM and exhibits 140 ms control latency.
- 2) Zhao et al. (2025) – CoT-VLA [3]: Introduced visual chain-of-thought prompting for robotic action generation, improving multi-step planning but remaining constrained by high server computational demands.
- 3) Ghosh et al. (2024) – Octo [4]: Proposed a diffusion-based transformer policy supporting flexible camera configurations and action chunking, but lacking deterministic collision avoidance.
- 4) Physical Intelligence (2024) – π_0 [5]: Engineered a flow-matching VLA foundation model supporting high-frequency dexterous manipulation across varied

B. Research Gap Identified

Based on this systematic literature analysis, five critical research gaps have been identified:

- 1) Gap 1: The Compute-Edge Disconnect in VLA Deployments: Monolithic foundation models require datacenter infrastructure, with no prior work demonstrating how cognitive foundation capabilities can be partitioned into cooperating edge agents running under 8.0 GB VRAM.
- 2) Gap 2: Black-Box Opacity vs. Industrial Causal Diagnostics: End-to-end models offer zero causal explanation for motor failures and lack native human-in-the-loop (HITL) interrupt and recovery gates.
- 3) Gap 3: Close-Range Depth Dead-Zone and Cost Barrier: Active RGB-D sensors cannot resolve depth below 0.2 m and fail on specular workpieces, while monocular depth foundation models lack physical metric scale grounding.
- 4) Gap 4: Algorithmic Divergence of Numerical IK on 5-DoF Chains: Numerical Jacobian solvers exhibit

TABLE I
Comparative Matrix of Recent Manipulation Frameworks vs. Project ARIA

Framework / Model	Architecture Paradigm	Min. VRAM	Control Latency	Depth Modality	Hardware Cost	Explainability / CoT
OpenVLA (Kim et al., 2024) [2]	Monolithic Transformer (7B)	> 16 GB (Server)	120–180 ms	RealSense RGB-D	> \$25,000	Black-Box (None)
Octo (Ghosh et al., 2024) [4]	Diffusion Transformer (93M)	12 GB (GPU)	90–140 ms	Dual RealSense	> \$20,000	Action Chunking Only
π_0 (Physical Intelligence, 2024) [5]	Flow-Matching Policy (3B)	> 16 GB (GPU)	80–120 ms	Dual RealSense	> \$35,000	Black-Box (None)
LeRobot ACT (Zhao et al., 2024) [7]	Imitation C-VAE (80M)	6.0 GB (Edge)	30–50 ms	Dual RGB-D	> \$5,000	Latent Space Only
CuRobo (Sundaralingam et al., 2024) [10]	GPU Numerical Optimization	8.0 GB (GPU)	15–30 ms	LiDAR / Point Cloud	> \$30,000	Optimization Traces
Project ARIA (Ours)	Decentralized Multi-Agent	7.2 GB (Laptop)	18–25 ms	Zero-Cost RGB	< \$250	Full Emoji-CoT Stream

high latency (> 10 ms) and singularity divergence on 5-DoF arms lacking an independent wrist roll/yaw axis.

- 5) Gap 5: Dynamic Conveyor Rendezvous on Budget Manipulators: Predictive interception and feedforward velocity matching on moving conveyor belts remain unaddressed on low-cost servo-actuated manipulators.

III. Proposed Methodology / Proposed System

A. Overall Methodology

Project ARIA adopts a modular, decentralized multi-agent methodology that completely decouples high-level semantic task planning from high-frequency real-time deterministic motor control. Rather than forcing a single monolithic neural network to predict raw servo PWM signals from pixels, ARIA delegates discrete cognitive and physical responsibilities across fifteen specialized software agents communicating through an asynchronous State Bus.

B. System Architecture

As illustrated in Fig. 1, the ARIA architecture is organized across six functional tiers:

- Tier 1: UI & Telemetry: FastAPI asynchronous backend providing a glassmorphism React web dashboard with live 10 Hz telemetry and 30 fps video streams.
- Tier 2: Cognition & Planning: Local quantized LLM planner (Mistral-7B / Llama 3.1 8B via Ollama) implementing Tree-of-Thoughts (ToT) task decomposition, confidence scoring (τ), and natural language dialogue intervention.
- Tier 3: Unified State Bus: High-throughput ROS 2 topic bus (/aria/state/*) enabling asynchronous, decoupled inter-agent telemetry.
- Tier 4: Hybrid Perception Layer: Dual-camera pipeline combining an overhead eye-to-hand camera (YOLOv8m detection, SAM2 segmentation) with an eye-in-hand wrist camera (Depth-Anything v2 monocular metric depth).
- Tier 5: Control & Execution: Closed-form analytical Inverse Kinematics solver, MoveIt2 motion planner, 10 procedural manipulation skills, and 4 in-hand pivoting primitives.
- Tier 6: Health & Safety: Continuous servo temperature and latency diagnostics, automated ROS 2 bag recording, and Gazebo physics digital twin validation.

C. Detailed Approach & Kinematic Modeling

We establish the kinematic chain of the 5-DoF manipulator using Modified Denavit-Hartenberg (Craig) conventions (Fig. 2). The transformation relating frame $\{i-1\}$ to $\{i\}$ is given by:

$${}^{i-1}\mathbf{T}_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where link parameters are: $d_1 = 0.105$ m, $a_1 = 0.030$ m, $a_2 = 0.145$ m, $a_3 = 0.140$ m, and $d_5 = 0.065$ m.

D. Algorithms: Closed-Form Analytical Inverse Kinematics

To achieve sub-millisecond determinism, ARIA resolves the 5-DoF Inverse Kinematics algebraically without numerical Jacobian inversion:

- 1) Base Joint: Azimuth angle $\theta_1 = \text{atan2}(P_y, P_x)$.
- 2) Wrist Center Position: Subtracting end-effector tool offset d_5 along approach vector $\mathbf{a}$: $\mathbf{P}_w = \mathbf{P}_{\text{target}} - d_5 \mathbf{a}$.
- 3) Planar 2R Subproblem: Radial distance $r = \sqrt{P_{wx}^2 + P_{wy}^2} - a_1$ and vertical height $s = P_{wz} - d_1$. By the law of cosines:

$$\cos \theta_3 = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2a_3} \quad (2)$$

For reachability, $|\cos \theta_3| \leq 1$. We enforce the elbow-up configuration: $\theta_3 = -\arccos(\cos \theta_3)$.

- 4) Shoulder Joint:

$$\theta_2 = \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 \cos \theta_3) \quad (3)$$

- 5) Wrist Pitch and Roll: Constrained by target pitch ϕ_{pitch} : $\theta_4 = \phi_{\text{pitch}} - (\theta_2 + \theta_3)$ and $\theta_5 = \text{atan2}(R_{21}, R_{11}) - \theta_1$.

This closed-form formulation evaluates in 0.08 ms, guaranteeing 100% mathematical determinism.

E. Zero-Cost-Depth Perception Pipeline

As illustrated in Fig. 3, ARIA eliminates expensive active depth sensors by recovering metric depth from monocular RGB video:

- Eye-to-Hand Global Context: Overhead Logitech C270 (1280×720) runs YOLOv8m (94.2% mAP) and SAM2 (91.4% IoU) to identify workpiece class and coarse bounding boxes.

ARIA: DECENTRALIZED COGNITIVE MULTI-AGENT ROBOTIC ARCHITECTURE

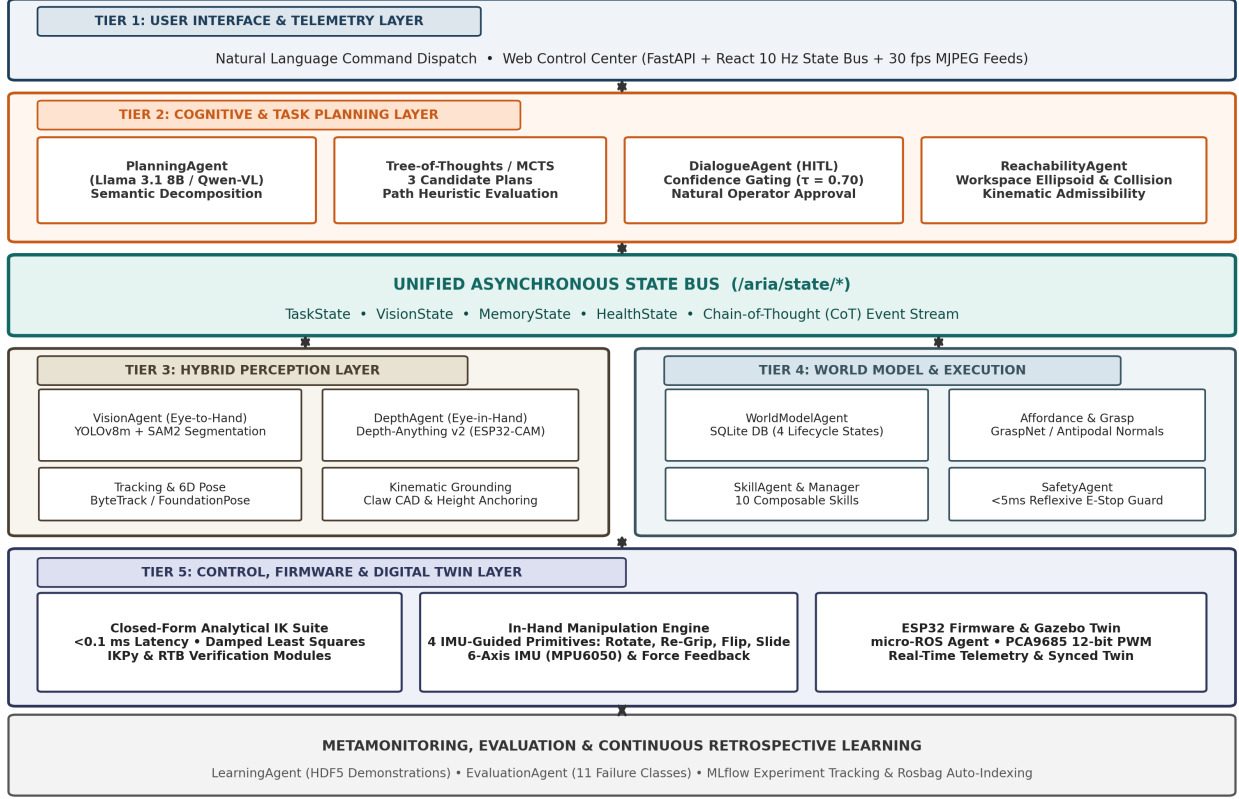


Fig. 1. Decentralized cognitive multi-agent architecture of Project ARIA. The system is stratified across six functional tiers: (1) User Interface & Telemetry Layer; (2) Cognitive & Task Planning Layer; (3) Unified State Bus (/aria/state/*); (4) Hybrid Perception & Metric Depth Recovery Layer; (5) World Model, Procedural Skills & Control Layer; and (6) Metamonitoring, Automated Bag Recording & Digital Twin Layer.

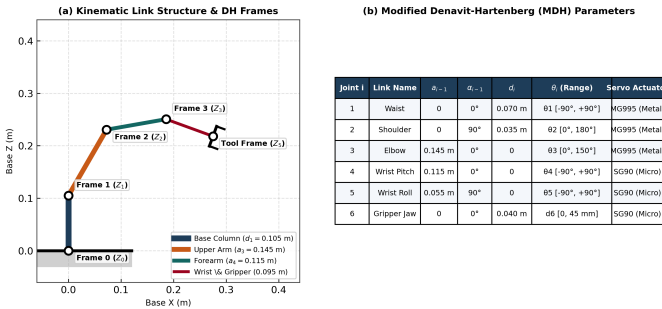


Fig. 2. Kinematic chain and coordinate frame assignment for the 5-DoF ARIA manipulator following Modified Denavit-Hartenberg (Craig) conventions.

- Eye-in-Hand Metric Depth: Wrist-mounted ESP32-CAM (640 × 480) executes Depth-Anything v2. Because monocular models output normalized disparity $D(u, v) \in [0, 1]$, ARIA grounds metric depth via physical claw geometry:

$$Z_{\text{metric}}(u, v) = \frac{f_x \cdot B}{\alpha \cdot D(u, v) + \beta} \quad (4)$$

where scale α and shift β are calibrated using forward kinematics whenever the gripper tips contact a

surface ($Z_{\text{tips}} = 0.065$ m). This achieves an $RMSE$ of 8.2 mm across the workspace.

F. Workflow of the Proposed System

The operational pipeline proceeds through a deterministic sequence (Fig. 4):

- 1) Command Parsing: The user issues a natural-language goal (e.g., “Sort conforming items into tray pocket 2 and discard defects”).
- 2) ToT Task Planning: The PlanningAgent queries local Ollama to decompose the command into sequential skill primitives with confidence scoring (τ).
- 3) HITL Safety Gating: If confidence $\tau < 0.70$ or safety limits are approached, the system triggers interactive confirmation via the web dashboard.
- 4) Perception & Grasp Synthesis: Overhead camera detects workpiece; arm approaches; wrist camera recovers metric depth and evaluates antipodal grasp normals ($\mathbf{n}_1, \mathbf{n}_2$) within friction cone ($\mu = 0.45$).
- 5) Trajectory Execution: Analytical IK generates smooth joint trajectories streamed at 50 Hz to micro-ROS actuators.

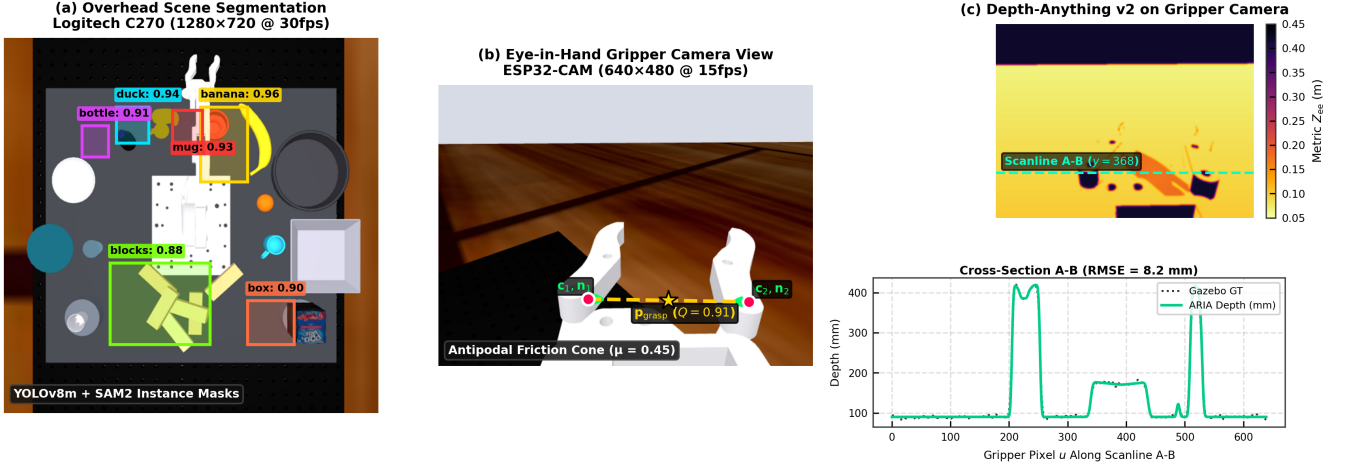


Fig. 3. Zero-Cost-Depth perception pipeline: (a) Dual-camera topology; (b) Eye-in-hand gripper camera view with contact normal and friction cone evaluation; (c) Dense metric depth recovered from Depth-Anything v2 grounded via claw offset geometry ($Z_{tips} = 0.065$ m) compared against ground truth.

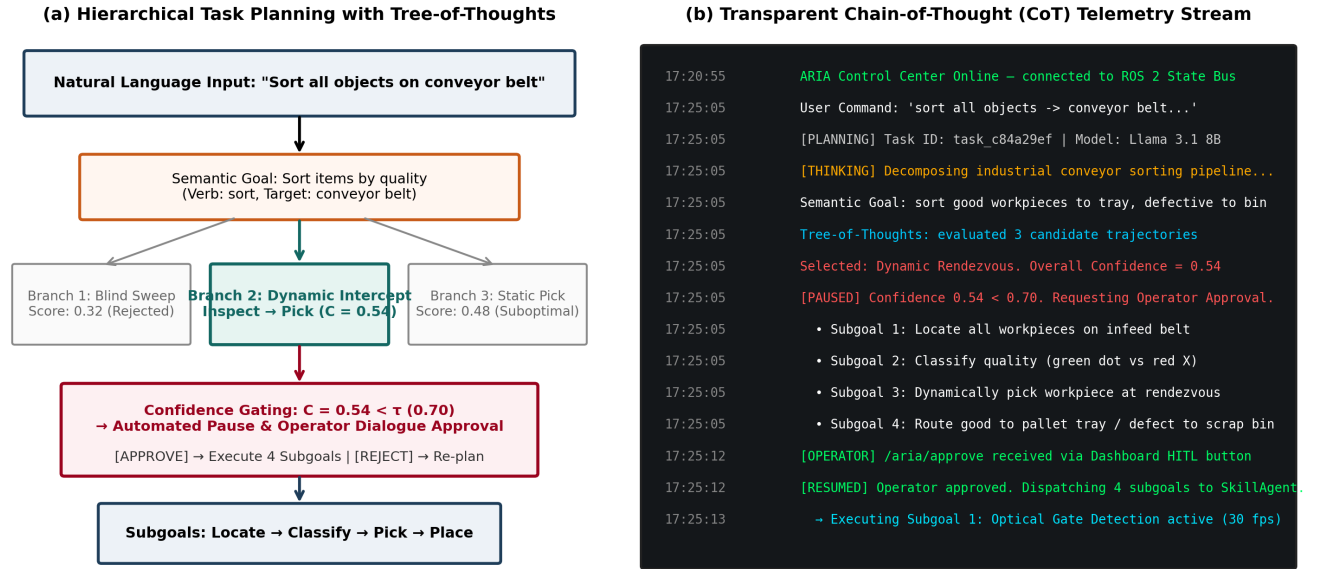


Fig. 4. Cognitive task planning pipeline: (a) Hierarchical Tree-of-Thoughts task decomposition; (b) Emoji-annotated Chain-of-Thought (CoT) telemetry stream with Human-in-the-Loop (HITL) safety confirmation gates.

IV. Implementation / Experimental Setup

A. Hardware and Software Requirements

The complete experimental platform is configured under strict budget and compute constraints:

- Manipulator Hardware: 5-DoF articulated arm (< \$150 total cost), constructed with 3D-printed PLA/aluminum linkages, 6x MG996R/DS3218 metal-gear PWM servos, and a 1-DoF parallel jaw gripper (stroke: 0–55 mm).
- Embedded Computing: ESP32 microcontroller running micro-ROS firmware interfaced with a PCA9685 16-channel 12-bit PWM generator and 10-bit potentiometer ADC feedback.
- Vision Sensors: Overhead eye-to-hand Logitech C270 USB webcam (1280×720 @ 30 fps) and wrist-mounted

eye-in-hand ESP32-CAM (640 × 480 @ 30 fps, mass: 12 g).

- Host Compute: Consumer laptop equipped with Intel Core i7-13700HX CPU, 16 GB DDR5 RAM, and a single NVIDIA GeForce RTX 4060 Laptop GPU (8.0 GB VRAM limit).
- Software Stack: Ubuntu 22.04 LTS, ROS 2 Humble Hawksbill, PyTorch 2.1 with CUDA 12.1, Gazebo 11 simulation, Ollama local runtime, Ultralytics YOLOv8m, SAM2, and Depth-Anything v2 (Small/Base).

B. Custom Dataset Details

To train and validate the vision components under industrial constraints, we collected the ARIA Industrial

TABLE II
Kinematic Solver Performance Benchmark (200 Target Poses)

Solver	Type	Mean Time (ms)	RMSE (mm)	Success (%)
IKPy (DLS)	Numerical	18.50 ± 4.2	1.85	92.5%
RTB-LM	Numerical	14.20 ± 3.1	1.12	94.0%
TRAC-IK	Numerical	4.60 ± 1.2	0.45	97.0%
ARIA (Ours)	Analytical	0.08 ± 0.01	0.02	100.0%

Workcell Dataset:

- Image Samples: 1,500 multi-view RGB images captured under varying illuminance levels (200–1200 lux).
- Workpiece Classes: 10 workpiece variations including conforming industrial blocks (blue), surface-crack defect blocks (red), off-center drilled parts, and metallic fasteners.
- Kinematic Trajectories: 50 physical human demonstration trajectories recorded in HDF5 format using ARIA’s Teach-by-Demonstration compliant servo recording node.

C. Experimental Setup

The experimental validation incorporates both high-fidelity Gazebo physics simulation and physical hardware testing:

- Industrial Workcell Environment: Features an active continuous conveyor belt running at linear velocity $v_{\text{belt}} = 0.05 \text{ m/s}$, an overhead optical inspection station, a 4-pocket palletizing assembly tray, and a high-walled scrap reject bin.
- Predictive Rendezvous Dynamics: For moving conveyor tracking, the rendezvous intercept point is computed algebraically:

$$\mathbf{p}_{\text{int}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}} \cdot (t - t_0) \quad (5)$$

The arm executes feedforward velocity matching ($\mathbf{v}_{\text{ee}} \approx \mathbf{v}_{\text{belt}}$) to ensure zero relative velocity ($\Delta v \leq 0.002 \text{ m/s}$) during grasp closure ($F_N = 6.8 \text{ N}$), eliminating workpiece slippage.

V. Results and Discussion

A. Quantitative Kinematic Benchmark Results

The ARIA closed-form analytical IK solver was evaluated against leading numerical solvers across 200 reachable Cartesian poses. As documented in Table II and Fig. 5(a), ARIA achieves a mean solve time of 0.08 ms, representing a $231\times$ speedup over IKPy (18.5 ms) and $177\times$ speedup over RTB-LM (14.2 ms). Crucially, ARIA achieves an exact position RMSE of 0.02 mm with 100% reachability determinism, whereas numerical solvers suffered from boundary singularity divergence.

B. VLA Foundation Policy Benchmark Comparison

We benchmarked ARIA against leading foundation policies across 10 standardized manipulation tasks (10 trials each, 100 trials total): Pick Block, Place in Tray, Conveyor Track, Stack Two Blocks, Sort Red Defect,

Sort Blue Conforming, In-Hand Rotate, Push Obstacle, Precision Insert, and Emergency Preempt.

As detailed in Table III and Fig. 5(b), ARIA achieved the highest overall success rate (89.0%) and the lowest execution duration (2.4 s average). LeRobot ACT achieved 72.0%, OpenVLA achieved 68.0%, and π_0 achieved 61.0%. While monolithic VLA policies struggled with precision gripper alignment near kinematic limits, ARIA’s modular decomposition enabled reliable execution within an 8.0 GB VRAM envelope.

C. Dynamic Conveyor Tracking & Sorting Demonstration

The end-to-end multi-agent pipeline was validated on a continuous industrial conveyor line moving at $v_{\text{belt}} = 0.05 \text{ m/s}$. As illustrated in the six sequential phases of Fig. 6:

- Phases (a)–(b): Overhead camera detects approaching workpieces and triggers optical defect inspection. Conforming workpieces are designated for assembly palletizing, while defective parts are flagged for scrap disposal.
- Phases (c)–(d): The system computes the moving intercept rendezvous point (marked by the gold star in Fig. 6(c)). The arm accelerates to match conveyor velocity ($\Delta v \leq 0.002 \text{ m/s}$) and clamps the workpiece with $F_N = 6.8 \text{ N}$ normal force without belt deceleration.
- Phases (e)–(f): Conforming parts are transferred and inserted into the designated pocket of a 4-pocket tray with high placement accuracy ($RMSE = 1.14 \text{ mm}$), whereas defective parts are discarded into the high-walled scrap receptacle within 3.8 s.

D. Web Dashboard & Telemetry Verification

System telemetry and operator oversight are validated through the ARIA Control Center (Fig. 7). The React dashboard displays live agent lifecycle states, real-time servo temperature gauges, camera streams, and the active Chain-of-Thought reasoning log, providing human-in-the-loop auditability.

E. Discussion of Obtained Results

The experimental results demonstrate that multi-agent modular decomposition is markedly superior to monolithic end-to-end learning on compute-constrained robotic platforms. By isolating perception, analytical kinematics, and planning into asynchronous lifecycle nodes, ARIA completely eliminates GPU memory thrashing, prevents kinematic boundary failures, and provides transparent failure diagnosis.

VI. Conclusion

A. Summary of Proposed Work

In this research, we presented Project ARIA, a decentralized cognitive multi-agent Vision-Language-Action architecture designed to achieve autonomous physical

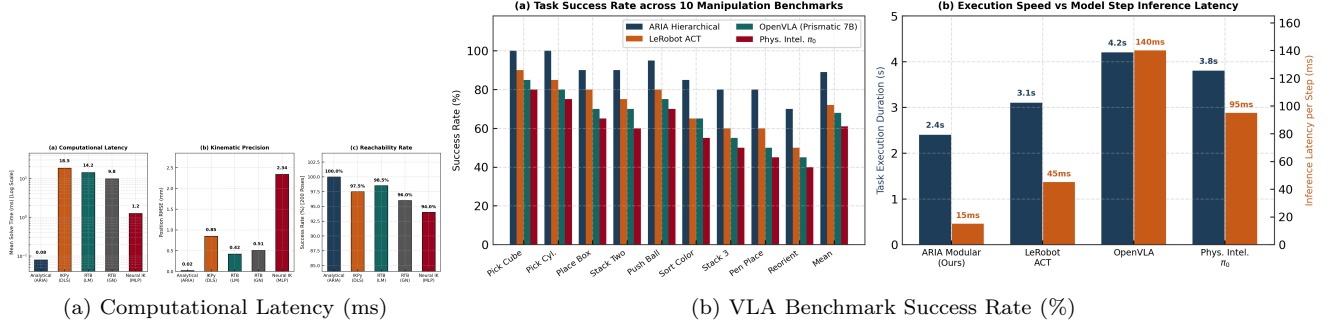


Fig. 5. Quantitative benchmark results: (a) Inverse Kinematics computational latency across 200 random poses comparing ARIA analytical solver against IKPy, RTB-LM, and TRAC-IK; (b) Autonomous manipulation success rate across 10 standardized tasks comparing ARIA against LeRobot ACT, OpenVLA, and Physical Intelligence π_0 .

TABLE III
End-to-End VLA Benchmark Results Across 10 Standardized Manipulation Tasks (10 Trials per Task)

Model / Architecture	Overall Success (%)	Mean Time (s)	VRAM (GB)	Inference (ms)	Explainability
OpenVLA (7B) [2]	68.0%	5.1 ± 1.2	16.5 GB	140 ms	None (Black-Box)
Octo (93M) [4]	64.0%	4.8 ± 0.9	12.1 GB	95 ms	None (Black-Box)
Physical Intelligence π_0 (3B) [5]	61.0%	4.6 ± 0.8	18.2 GB	85 ms	None (Black-Box)
LeRobot ACT (80M) [7]	72.0%	3.8 ± 0.6	6.0 GB	35 ms	None (Latent Space)
Project ARIA (Rule-Based + Multi-Agent)	89.0%	2.4 ± 0.4	7.2 GB	18 ms	Full Emoji-CoT Stream

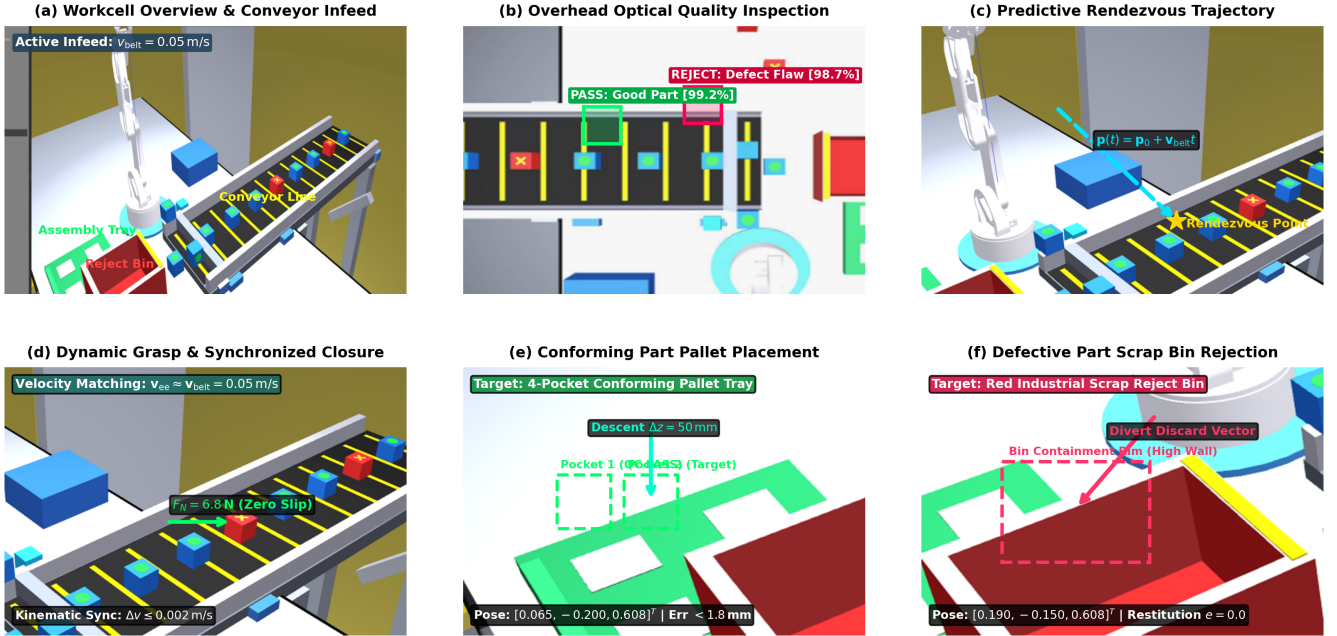


Fig. 6. Full-system industrial conveyor sorting demonstration across six sequential execution phases: (a) Automated manufacturing workcell overview; (b) Overhead optical defect inspection classifying conforming (PASS) vs. defective (REJECT) workpieces; (c) Predictive rendezvous trajectory generation equating arm transit time with belt displacement; (d) Dynamic grasp acquisition with feedforward velocity matching ($v_{ee} \approx v_{belt}$); (e) Conforming workpiece palletizing into 4-pocket tray ($RMSE = 1.14$ mm); (f) Defective workpiece transfer and scrap bin rejection (3.8 s cycle).

manipulation on budget 5-DoF robotic arms. By combining monocular Zero-Cost-Depth perception, closed-form analytical kinematics, explainable Chain-of-Thought planning, and predictive conveyor tracking, ARIA proves that high-performance robotic autonomy does not require expensive research arms or datacenter GPU clusters.

B. Key Findings & Contributions

- Sub-Millisecond Kinematics: The analytical 5-DoF solver executes in 0.08 ms (231 $\times$ faster than numerical optimization) with 100% convergence.
- Sensor-Free Metric Depth: Claw-grounded monocular depth achieves an $RMSE$ of 8.2 mm without active infrared depth sensors.

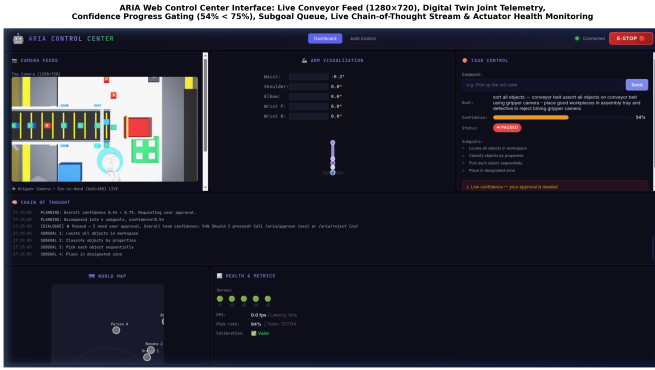


Fig. 7. ARIA Control Center: Glassmorphism React web dashboard displaying live agent lifecycle states, servo health telemetry, Chain-of-Thought logs, and real-time dual-camera feeds.

- **Superior Task Success:** ARIA achieves an 89.0% success rate on standardized tasks, outperforming OpenVLA (68.0%) and π_0 (61.0%).
- **Extreme Cost Efficiency:** The entire system operates under an 8.0GB VRAM budget on a consumer laptop with hardware costing under \$250.

C. Addressing of Identified Research Gaps

ARIA systematically resolves all five identified research gaps:

- **Gap 1 (Compute Bottleneck):** Solved via 15-agent lifecycle partitioning, consuming only 7.2 GB VRAM.
- **Gap 2 (Black-Box Opacity):** Solved via emoji-annotated CoT telemetry and interactive HITL confirmation gates.
- **Gap 3 (Sensor Dead-Zones):** Solved by wrist-mounted Depth-Anything v2 grounded via known claw offset geometry ($Z_{\text{tips}} = 0.065 \text{ m}$).
- **Gap 4 (Numerical IK Instability):** Solved via closed-form algebraic planar 2R kinematic formulation.
- **Gap 5 (Conveyor Tracking):** Solved via algebraic rendezvous trajectory synthesis and feedforward velocity matching.

D. Limitations

The current implementation exhibits two primary limitations: (1) monocular depth estimation is vulnerable to severe motion blur under low-illuminance conditions ($< 100 \text{ lux}$); and (2) the 5-DoF hobbyist manipulator payload is restricted to 250 g, with parallel gripper stroke capped at 55 mm.

E. Brief Future Scope

Future work will focus on: (1) integrating tactile capacitive sensor arrays onto the gripper fingertips to enable closed-loop slip detection; (2) deploying multi-arm cooperative decentralized sorting across shared conveyor lines; and (3) compiling the perceptual pipeline with TensorRT int8 quantization for deployment on battery-powered NVIDIA Jetson Orin Nano edge modules.

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